



Autistic Children's Processing of Emotional Prosody: A Native Language Advantage

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Abstract

Purpose The ability to understand emotional prosody is directly relevant to effective social communication. Typically developing (TD) children at age 3 start to understand emotional prosody in both native and foreign languages, but it remains unclear whether autistic children demonstrate a similar pattern. To address this, this study examined Mandarin-speaking autistic children's understanding of two basic emotional prosody, happy and sad, across languages.

Methods We manipulated the language where emotional prosody was presented: familiar (Mandarin) versus unfamiliar (French and Spanish). Autistic 3- and 5-year-olds and age-matched TD children completed a face-sound matching task where they were asked to point at the emotional face that matched the spoken sentence they heard.

Results Autistic children exhibited similar patterns to age-matched TD peers in the native language condition but showed significantly lower accuracy when responding to emotional prosody in foreign languages. Importantly, however, autistic children's performance across all language conditions was above chance, and within-group analysis showed that they performed better in the native language condition than in the foreign language conditions.

Conclusion These findings suggest that autistic children can decode emotional prosody across languages, although their ability to process emotional prosody in foreign languages is less developed than TD children. This pattern might reflect fundamentally different mechanisms between the two groups. Alternatively, it could also be explained by shared mechanisms between the two groups, with the difference reflecting autism-related neurocognitive characteristics constraining autistic children's processing efficiency. In addition, the native language advantage can be leveraged to support emotional communication in autistic children.

Keywords Emotional prosody · Autistic children · Cross-linguistic comparison · Generalization · Language input

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Introduction

Prosody refers to the suprasegmental characteristics of speech deriving from variations in duration, amplitude, and fundamental frequency of speech sounds. The prosodic patterns of spoken sentences often signal a speaker's emotional and mental states, for example, whether the speaker is happy or sad, or whether s/he is making a statement or asking a question (Ladd, 1996; Prieto, 2015; Wilson & Wharton, 2006). We refer to its function to convey emotional information as emotional prosody. The ability to understand emotional prosody is directly relevant to effective social communication (Durkin & Conti-Ramsden, 2007; Foxe et al., 2015). Typically developing (TD) children at age 3 start to reliably understand emotional prosody in their native language (Baltaxe, 1991; Stifter & Fox, 1986) and in foreign languages (Ma et al., 2022). The ability to decode emotional

prosody continues to develop during preschool years up to 9 or 10 years of age (Chronaki et al., 2015, 2018; Gil et al., 2016; Nelson & Russell, 2011; Sauter et al., 2013).

Compared with their TD peers, autistic children exhibited more difficulties in understanding basic and complex emotions conveyed by prosodic information (Boucher et al., 2000; Chevallier et al., 2011; Fridenson-Hayo et al., 2016; Lartseva et al., 2015; Le Sourn-Bissaoui et al., 2013; Paul et al., 2005; Peppé et al., 2007; Scheerer et al., 2020; Song et al., 2020; Wang & Tsao, 2015).¹ For example, Scheerer et al. (2020) investigated the ability to extract basic emotions (e.g., happiness, sadness, anger, and surprise) from nonverbal emotional voices by 6- to 13-year-old autistic children. In the study, the participants were asked to match an emotional voice to an emotional face or an emotional word. The results showed that autistic children had no trouble matching emotional voices to the corresponding emotional words, but they exhibited great difficulty when matching emotional voices to emotional faces. According to Scheerer et al., the finding that autistic children were able to match emotional voice to emotion words might suggest that, under some circumstances, they can accurately extract the emotional meaning conveyed by prosody, but at the same time, they had difficulty in matching emotional voice to emotional faces, which might suggest that they may struggle to make use of prosody in a social context. Similar findings were reported in autistic children between the ages of 6 and 10 by Boucher et al. (2000).

In addition, while difficulties in extracting the emotional meaning from prosody are frequently reported for autistic children, there is also evidence to suggest that the ability to extract emotion from prosody is intact in autistic individuals (Begeer et al., 2008; Brennard et al., 2011; Loveland et al., 1997; Prior et al., 1990). For example, Begeer et al. (2008) found that high-functioning autistic children were able to recognize emotions when explicit cues were provided, e.g., selecting pictures of facial expressions, naming emotions, or responding to explicit expressions of emotions. According to Begeer et al., autistic children's responses were guided by a relatively later-acquired, explicit understanding of emotions, which was highly dependent on intellectual abilities. In other words, autistic children might develop compensatory strategies by explicitly learning the relation between an emotional label and its corresponding prosodic information.

¹ It should be noted that autistic children in these studies were required to decode neurotypical emotional prosody, as the experimental stimuli were produced by neurotypical individuals in neurotypical contexts, rather than emotional prosody produced by autistic individuals in autistic communicative contexts. As pointed out by an anonymous reviewer, the observed difficulties of the autistic children in these studies might be a reflection of being in a neurotypical environment, where autistic individuals are evaluated by neurotypical standards.

Although previous research has yielded mixed results, an emerging pattern is that autistic children might differ from TD children in the way emotional prosody is processed across contexts. Autistic children seem able to respond accurately to emotional prosody in explicit contexts, but their learned skills are often not sufficient to ensure accurate responses in natural contexts. This might partially explain the discrepancy in previous findings. TD children can establish abstract mapping relations between prosody and emotion and can spontaneously use this knowledge in natural contexts, whereas autistic children rely on compensatory strategies, and their learned skills might be insufficient for them to spontaneously use the skills in natural contexts, and thus they often need an explicit context in which cues are clearly provided. This difference between autistic children and their TD peers raises interesting questions regarding the mapping knowledge of prosody and emotion in autistic children, i.e., what factors contribute to variations in how the mapping relations are learned and how flexibly they are applied across different contexts.

One possible explanation relates to the difference in generalization abilities between autistic and TD children. Generalization abilities refer to the capacity to apply learned knowledge or skills beyond the context in which they were initially acquired, across different people, settings, materials, etc. (Stokes & Baer, 1977). Previous research suggests that autistic children face challenges in generalization (e.g., Church et al., 2015; de Marchena et al., 2015; Gastgeb & Strauss, 2012; Gunning et al., 2019; Naigles et al., 2013; Rincover & Koegel, 1975), and such challenges have been documented not only in intervention settings, where skills learned in structured training often fail to transfer to everyday life, but also in non-intervention contexts. For example, de Marchena et al. (2015) examined autistic children's ability to transfer a strategy (i.e., mutual exclusivity) used in one context to a similar but different context and found that autistic children were less likely to learn from previous experience, and their generalization was less consistent than their TD peers, indicating generalization weakness in non-intervention contexts. These findings suggest that difficulties in generalization might explain why autistic children respond accurately to emotional prosody in explicit tasks but face challenges in more natural contexts that require flexible use of their knowledge about the mapping relations between prosody and emotion.

TD children's understanding of emotional prosody generalizes not only across contexts but also across languages, as previous research has shown that TD children as young as three years old can understand emotional prosody in their native language (Baltaxe, 1991; Stifter & Fox, 1986) as well as in foreign languages (Ma et al., 2022). The mapping relation between prosody and emotion is often considered

to be cross-linguistically valid, and listeners rely on prosodic cues, sometimes more than lexical meaning, when interpreting the emotional meaning of an utterance (Mehrabian, 1968). However, it remains unclear whether autistic children can process emotional prosody across languages. Given that emotional prosody is typically presented in their native language and that autistic children may experience difficulties with generalization, it is highly likely that they are able to understand emotional prosody in their native language but experience difficulties when it is expressed in an unfamiliar foreign language.

Therefore, the present study aims to examine autistic children's ability to decode emotional prosody in both their native language and foreign languages to see whether their understanding of emotional prosody is cross-linguistic or more language-specific. Addressing this question will not only shed light on the mechanisms underlying the processing of emotional prosody in autistic children but also contribute to our broad understanding of their language and cognitive development.

Methods

Participants

Age range selection The present study focused on preschool autistic children. Three-year-old autistic children were selected as the youngest group based on prior research showing that TD children at the age of 3 start to reliably understand emotional prosody both in their native language and in foreign languages (e.g., Baltaxe, 1991; Ma et al., 2022; Stifter & Fox, 1986). Including this age group, therefore, allowed us to examine whether autistic children show comparable early sensitivity to emotional prosody. Five-year-olds were included to explore the developmental pattern of children's ability to map prosody onto emotion.

Three- and five-year-old TD children were included to provide both age-matched and verbal-IQ-matched comparisons. First, TD children in each age group were age-matched with autistic 3- and 5-year-olds to examine whether autistic and TD children of the same chronological age show similar understanding of emotional prosody. Second, TD 3-year-olds served as the verbal-IQ-matched comparison for autistic 5-year-olds. This comparison was included to determine whether autistic and TD children with comparable verbal ability differed in their understanding of emotional prosody, as the experimental task was language-related. Four-year-olds were not included in the present study in order to maintain a clear and consistent matching design across groups.

Participants' verbal IQ was operationalized as their Verbal Comprehension Index (VCI) standard score on the Wechsler Preschool and Primary Scale of Intelligence™ - IV (CN) (Li & Zhu, 2014). The VCI reflects children's acquired knowledge, verbal reasoning and comprehension skills, and the ability to attend to verbally presented stimuli. For 3-year-old children, the VCI was derived from the Receptive Vocabulary and Information subtests, and for 5-year-old children, it was derived from the Information and Similarities subtests. Raw scores on these subtests were converted to VCI standard scores according to the WPPSI-IV manual.

Sample size calculation Minimum sample sizes for each condition ($n=25$) were established by conducting a power analysis using G*Power, based on a medium expected effect size ($f = 0.25$) and the use of a repeated-measures analysis of variance containing four groups (two age groups of autistic children and two age groups of TD children herein) and three measures (three languages, Mandarin Chinese, French, and Spanish herein; Faul et al., 2007). This minimal sample size was based on prior studies on children's understanding of emotion in speech and music (e.g., Ma et al., 2022; Morton & Trehub, 2001; Mote, 2011; Nawrot, 2003; Quam & Swingley, 2012). To maximize statistical power, we permitted modest increases to sample sizes according to the recruitment capacity for each group.

Autistic group Fifty-eight autistic 3-year-olds and 62 autistic 5-year-olds participated in the study. They were all monolingual Mandarin speakers in Beijing, China, and were recruited from the Enqi Autism Platform in Beijing. None of them had learned French or Spanish. Their diagnoses were made by pediatric neurologists at hospitals using DSM-5 (American Psychiatric Association, 2013). To further confirm the participants' diagnoses, each participant was independently evaluated using the Autism Diagnostic Observation Schedule, Second Edition (ADOS-2; Lord et al., 2012) by two members of our research team who have the qualification of administering and coding the ADOS-2 for clinical and research reliability. In the current study, Module 2 was selected based on children's language and developmental levels. Four of the 3-year-olds and two of the 5-year-olds were excluded because they did not meet the autism cut-off on the ADOS-2 (total score ≥ 10 for the 3-year-olds, total score ≥ 9 for the 5-year-olds). Four additional children did not complete the study, including two 3-year-olds and two 5-year-olds. The final sample was comprised of 52 autistic 3-year-olds (mean age 3;6, range 3;5–3;9, 22 girls) and 58 autistic 5-year-olds (mean age 5;7, range 5;5–5;9, 26 girls). All the autistic participants had a verbal IQ score above 80.

TD comparison group Sixty-six TD 3-year-olds and 80 TD 5-year-olds were recruited from Beijing, China, and

Table 1 Verbal IQ scores of each participant group (SD in parentheses)

Group	Number	Verbal IQ
Autistic groups		
3-year-olds	52	92.33 (11.56)
5-year-olds	58	101.17 (11.21)
TD groups		
3-year-olds	64	101.28 (11.05)
5-year-olds	80	105.72 (10.46)

they were all monolingual Mandarin speakers. According to the parental reports, the participants had no history of clinical diagnosis, language disorder, or special educational services, and they also had no family history of diagnosed neurodevelopmental disorders. In addition, none of them had learned French or Spanish. The participants were also screened for potential undiagnosed autism by an expert clinician in our research team using DSM-5, and none of them met the DSM-5 criteria for autism. Two of the 3-year-olds were not included in the final analysis because they failed to complete the study. The final sample consisted of 64 TD 3-year-olds (mean age 3;5, range 3;3–3;9, 30 girls) and 80 TD 5-year-olds (mean age 5;6, range 5;2–5;9, 40 girls). All the TD participants had a verbal IQ score above 80.

Groups and matching As aforementioned, we matched the autistic and TD participants on both age and verbal IQs. Table 1 shows the descriptive characteristics of the two autistic groups and the two TD groups divided by age. The autistic 5-year-olds were matched with the TD 3-year-olds on verbal IQ ($t(120)=0.56$, $p=.28$, Cohen's $d=0.01$). Compared with their age-matched TD peers, the autistic 5-year-olds had significantly lower verbal IQ scores ($t(136)=4.77$, $p<.01$, Cohen's $d=0.42$). In addition, compared with the TD 3-year-olds, the autistic 3-year-olds had significantly lower verbal IQ scores ($t(114)=5.89$, $p<.01$, Cohen's $d=0.79$).

All autistic and TD children were from middle-class families in Beijing, with no systematic differences in family background within and between the two groups.

The study was approved by the Ethics Committee of the School of International Studies, Zhejiang University, SIS2023-04. Written informed consent has been obtained from each participant's parent.

Materials and Design

The present study was modelled closely on the task by Ma et al. (2022) and used similar auditory and visual stimuli as in the happy-sad condition by Ma et al. (2022). Happiness and sadness were chosen because they are widely regarded as basic emotions across cultures (Ekman, 1992; Oatley & Johnson-Laird, 1987; Shaver et al., 1987) and have been extensively used in prior empirical studies of emotion recognition in children (e.g., Baltaxe, 1991; Kastner & Crowder, 1990; Stifter & Fox, 1986). Both the auditory and visual stimuli have been validated in Ma et al. (2022). Importantly, we manipulated language familiarity: familiar (Mandarin, the native language to which the participants had sufficient exposure) vs. unfamiliar (French and Spanish, two foreign languages to which the participants had no exposure). More specifically, two independent variables were manipulated: emotional prosody (happy and sad) and the language in which the emotional prosody was presented (Mandarin, French, and Spanish).

Female native speakers of Mandarin, French, and Spanish produced four test sentences in their respective native languages (see Table 2 for the four test sentences in the three languages), which were selected from a database of semantically neutral sentences (Ben-David et al., 2011). Each sentence was produced to convey two emotional connotations: happy and sad, leading to 24 test sentences in total. The speakers were instructed to produce the sentences in a natural manner while explicitly conveying the intended emotions.

In addition, 12 pairs of line-drawn schematic emotional facial expressions that corresponded to happy and sad emotions were also created. Example emotional facial expressions are provided in Fig. 1. In each pair, the two facial expressions differed only in the shape of the eyebrows, eyes, and mouths, which served as the critical cues distinguishing happiness from sadness. Line-drawn schematic emotional faces are often used to test 3- to 5-year-olds' understanding of emotion in prior research (e.g., Dalla Bella et al., 2001; Kastner & Crowder, 1990), and they were successfully used

Table 2 Test sentences used in the study with the English translations in the parenthesis

Mandarin	French	Spanish
我看到地上的毯子 (‘I see a rug on the floor.’)	Je vois un tapis au sol	Veó un tapete en el piso
这是个垃圾桶 (‘This is a garbage can.’)	Ceci est un poubelle	Es un basurero
我的勺子在桌上 (‘My spoon is on the table.’)	Ma cuillère est sure la table	Mi cuchara esta en la mesa
毛巾折过了 (‘One towel is folded.’)	Une serviette est pliée	La toalla esta doblada

Two prosodic versions of each sentence were recorded, one expressing happy emotion and one expressing sad emotion

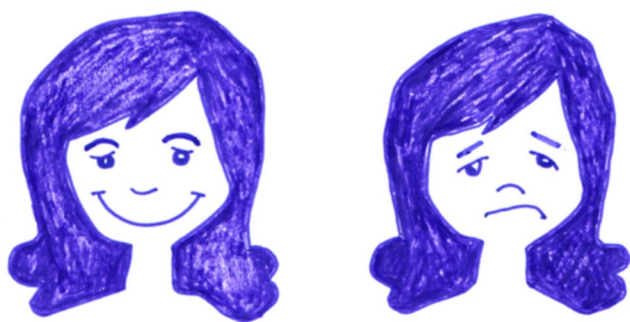


Fig. 1 Example emotional facial expressions in the study, happy face (left panel) and sad face (right panel)

in Ma et al. (2022) to test 3- to 5-year-old Mandarin-speaking children's understanding of emotional prosody.

In total, 12 test trials were constructed, each comprising a pair of happy and sad facial expressions (see Fig. 1 for an example) and a spoken sentence that was expressing either happy or sad emotion. Facial expressions were selected from the 12 pairs of line-drawn schematic emotional facial expressions, and spoken sentences were selected from the 24 test sentences that were expressing either happy or sad emotion in one of the three languages. The 12 test trials were divided into two experimental lists, with each participant seeing each pair of the facial expressions but listening to only one of two prosodic versions of each spoken sentence that were recorded for each language. Test sentences with happy emotional prosody and those with sad emotional prosody were counterbalanced across the two lists, with 6 in each prosodic condition for each list. In addition, test sentences in each of the three languages were counterbalanced. In each list, there were 4 test sentences in each of the three languages, including two sentences with happy prosody and two with sad prosody.

In each experimental list, the 12 test trials were arranged in random order. Participants in each group were randomly assigned to one of the two lists: 26 autistic 3-year-olds, 29 autistic 5-year-olds, 32 TD 3-year-olds and 40 TD 5-year-olds were assigned to List 1, and the other 26 autistic 3-year-olds, 29 autistic 5-year-olds, 32 TD 3-year-olds and 40 TD 5-year-olds were assigned to List 2.

Procedure

As in Ma et al. (2022), the participants were tested using a face-sound matching task, where they were asked to point at the line-drawn schematic emotional face that matched the spoken sentence they heard. Pairs of emotional faces as in Fig. 1 were shown to the participants side by side in a split-screen display on a laptop computer. The spoken sentences were presented to the participants through the same laptop connected to two external speakers. This task has been

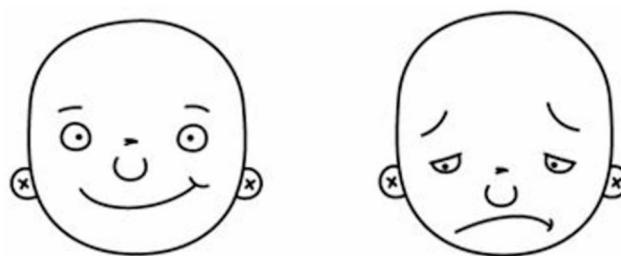


Fig. 2 Facial expressions on the practice trials, happy face (left panel) and sad face (right panel)

widely used to test preschool children's understanding of emotional sound (e.g., Dalla Bella et al., 2001; Kastner & Crowder, 1990). The autistic children were tested individually in a quiet room at a local training centre, and the TD children were tested individually in a quiet room at a local kindergarten.

Before the experiment started, the participants were instructed that they were going to play a game matching the emotion of someone's voice to a picture of an emotional face. A female experimenter also explained what emotion was by telling the participants, "An emotion is something we feel. For example, when we get to play a really fun game, we feel happy, but later when we have to stop playing that game, we feel sad. When we feel different emotions, we show these emotions to other people in different ways. For example, when we feel happy, we often smile a lot and our voice sounds very cheerful, but when we feel sad, we often frown, and our voice sounds gloomy." When talking about how the happy or sad voice sounded, the experimenter also imitated a cheerful or a gloomy voice.

After the instruction, the experiment started with two practice trials to familiarize the participants with the task. On the first practice trial, the participants were presented with two hand-drawn, schematic facial expressions (a happy face and a sad face of the same baby as in Fig. 2) side by side in a split-screen display connected to a laptop while listening to a giggling sound played through the external speaker connected to the same laptop. The experimenter then asked the participants, "Can you point at the face that matched the sound?" On the second practice trial, the participants saw the same facial expressions in Fig. 2 while listening to a crying sound, and they were asked to point at the face that matched the sound. All the participants responded correctly to the two practice trials, indicating that they were able to distinguish the happy and sad facial expressions and correctly map them onto the corresponding emotional sounds. The participants then proceeded to the actual test.

In the actual test, the participants were presented with 12 test trials. On each trial, they saw two emotional facial expressions as in Fig. 1 while hearing spoken sentences with either happy or sad emotional prosody. As on the practice

trials, the participants were instructed to point at the facial expression matching the given spoken sentence. On both the practice and test trials, no feedback was provided to the participants' responses. Pointing was used as the dependent measure, as it doesn't require a verbal response and thus can minimize the task demand posed on autistic children.

Results

We measured the participants' pointing accuracy. The mean accuracy of each participant group was summarized in Table 3. As shown in the table, the TD 5-year-olds correctly chose the facial expressions that matched the corresponding emotional prosody in both Mandarin and the two foreign languages. The TD 3-year-olds also performed well to emotional prosody in both Mandarin and the two foreign languages, but their performance was not as good as that of the TD 5-year-olds. The autistic 5-year-olds correctly chose the facial expressions that matched the corresponding emotional prosody in Mandarin, performing comparably to their TD peers. However, when emotional prosody was presented in the two foreign languages, their accuracy was significantly lower than that of age-matched and verbal-IQ-matched TD children. A similar response pattern was observed in the autistic 3-year-olds.

To assess the observed patterns statistically, we first fitted a maximal model afforded by the obtained data. We then simplified the maximal model to find the simplest model with an equivalent explanation power to the maximal model. Following the standard procedures, the model comparisons and simplifications were based on the model's likelihood ratio. More specifically, if two or more models' likelihood ratios were not significantly different, then the simple model was reported; otherwise, the complex model was used. We reported the results of the simplest best-fitting model. To justify the robustness of the observed effects, parametric bootstrapping was used with the best-fitting

model, in which we simulated a series of response vectors from the model using the estimated parameter values and then refitted the model to these responses in turn to accumulate the statistics of interest at each iteration. In total, 1000 of such interactions were generated and a ridge plot for the bootstrap samples of the fixed effects was then plotted. In addition, to better understand the observed effects, we then conducted a post-hoc examination of pairwise differences. All fitted models used the participants' response *accuracy* as the response variable, and the response variable was bounded between 0 and 1. All models were generalized linear mixed-effects models (GLMMs) with the Bernoulli distribution. The models were fitted with the package *MixedModels.jl* (Alday & Bates, 2025) in *Julia* programming language (Bezanson et al., 2017). The model comparisons and bootstrapping were conducted using the *likelihoodratio* function and the *parametricbootstrap* function respectively, in the package *MixedModels.jl*. The post-hoc analyses were conducted with *Effects.jl* (Alday et al., 2025) in *Julia* programming language. Graphical plots were rendered with *Makie.jl* (Danisch & Krumbiegel, 2021) in *Julia* programming language.

More specifically, our first analyses included three dummy-coded fixed terms: *Group* (autistic children vs. TD children, autistic children as the baseline), *Language* (Mandarin vs. French vs. Spanish, Mandarin as the baseline), and *Age* (3-year-olds vs. 5-year-olds, 3-year-olds as the baseline). The fixed effects included the three terms' main effects and their interactions. The model had one random term: *Subject*. The random effects included the random intercept for the random term and its slope with respect to the fixed term *Language*. The other two fixed terms *Group* and *Age* were between-participant variables and were not included in the slopes. The maximal model was $accuracy \sim 1 + Group * Language * Age + (1 + Language | Subject)$ in *Julia*. The simplest model was $accuracy \sim 1 + Group + Language + Age + Group \& Language + Group \& Age + Language \& Age + (1 | Subject)$. The fitting accuracies of the two models were similar ($\chi^2(2) = 0, p = .9705$). The three-way interaction of the three fixed effects was not significant. The model fitting results of the simplest best-fitting model are summarized in Table 4. As shown in the table, there was no significant effect of *Group* ($b = -0.01, z = -0.03, p = .9716$), indicating that the autistic 3-year-olds and the TD 3-year-olds exhibited similar performance when responding to the Mandarin emotional prosody. All other main effects and interactions were significant.

To justify the robustness of the observed effects, parametric bootstrapping was used with the best-fitting model, in which we simulated a series of response vectors from the model using the estimated parameter values and then refitted the model to these responses in turn to accumulate the

Table 3 Mean accuracy of each participant group in each language condition (standard deviation in parentheses)

Language	Group	Mean accuracy (SD)
Mandarin	TD 3-year-olds	89.45% (4.32%)
	TD 5-year-olds	99.24% (1.65%)
	Autistic 3-year-olds	89.37% (4.26%)
	Autistic 5-year-olds	97.28% (2.88%)
French	TD 3-year-olds	81.56% (5.09%)
	TD 5-year-olds	93.83% (4.30%)
	Autistic 3-year-olds	61.17% (6.89%)
	Autistic 5-year-olds	59.44% (4.66%)
Spanish	TD 3-year-olds	80.39% (5.01%)
	TD 5-year-olds	92.72% (3.99%)
	Autistic 3-year-olds	63.53% (4.25%)
	Autistic 5-year-olds	59.74% (5.10%)

Table 4 Statistical results of the simplest best-fitting model

Fixed effect	β	SE	z	Pr(> z)
(Intercept)	2.42	0.27	9.03	<1e-18***
Group: TD	-0.01	0.35	-0.03	0.9761
Language: French	-1.97	0.28	-7.00	<1e-11***
Language: Spanish	-1.86	0.29	-6.50	<1e-10***
Age: 5-year-olds	1.56	0.46	3.36	0.0008***
Group: TD & Language: French	1.05	0.37	2.86	0.0042**
Group: TD & Language: Spanish	0.90	0.37	2.43	0.0152*
Group: TD & Age: 5-year-olds	1.26	0.25	5.02	<1e-06***
Language: French & Age: 5-year-olds	-1.61	0.47	-3.44	0.0006***
Language: Spanish & Age: 5-year-olds	-1.72	0.47	-3.65	0.0003***

* $p < .05$, ** $p < .01$, *** $p < .001$

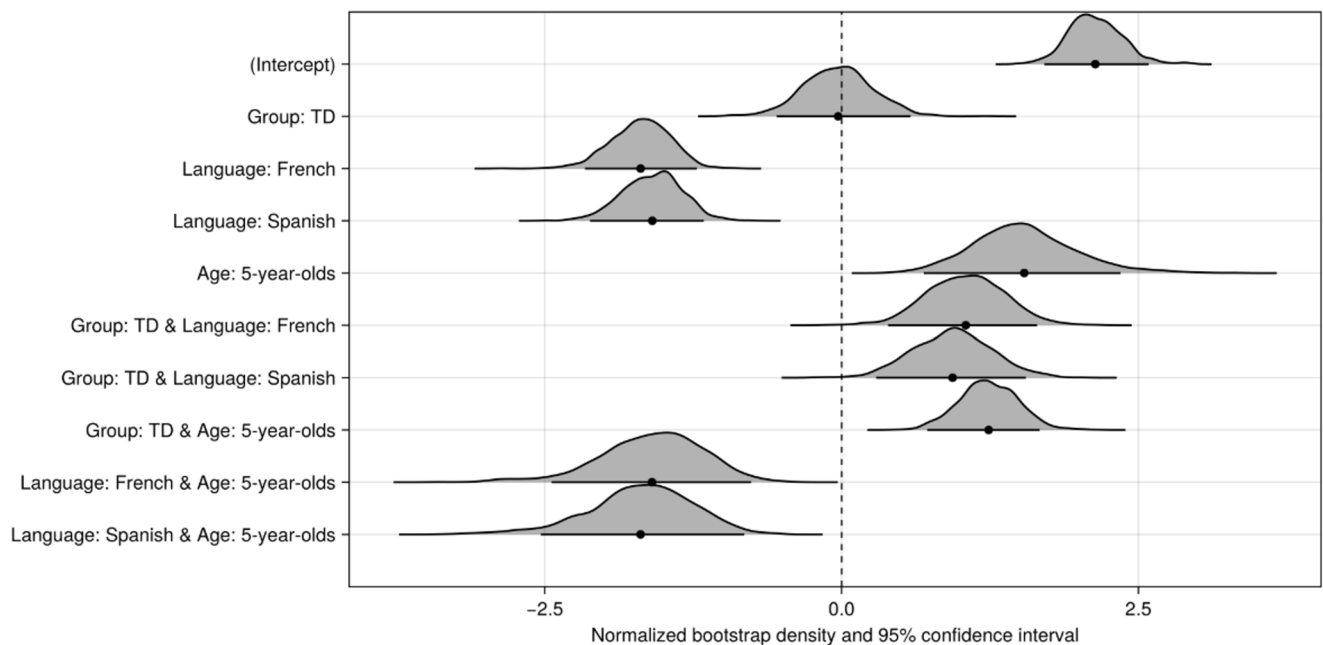


Fig. 3 The ridge plot for the bootstrap samples of the fixed effects for the simplest best-fitting model to the full data. Each row in the figure corresponds to the results of one specific fixed effect, consisting of the original estimated coefficient (the dot), the 95% confidence interval

(the line), and the density plot of the simulated values. The density plot and/or the confidence interval that do not contain the zero value suggests that the original estimate of the fixed effect was significantly robust

statistics of interest at each iteration. The results of the 1000 simulated iterations were displayed as a ridge plot in Fig. 3. As shown in the figure, all the significant effects observed in the original model fitting were robust in our simulated samples.

To explore these significant effects in detail, we then conducted a post-hoc analysis using the *empairs* function in *Effects.jl* (Alday et al., 2025). More specifically, we computed the pairwise differences of the estimated marginal means, which are summarized in Figs. 4, 5, and 6.

First, the differences between the autistic and TD children in each age group (3- and 5-year-olds) under each language condition (Mandarin, French and Spanish) are shown in Fig. 4. As indicated in the figure, the autistic and TD children responded equally well when the test language was

Mandarin, regardless of their age. However, when the test language was a foreign language (French or Spanish), the autistic children's accuracy was significantly lower than that of the TD children, regardless of their age.

Second, to explore the developmental patterns, the responses between the 3-year-olds and the 5-year-olds in both the autistic and TD groups under each language condition are provided in Fig. 5. As illustrated in the figure, the TD 5-year-olds performed significantly better than the TD 3-year-olds in all the three language conditions (see the left panel of Fig. 5). By contrast, no significant difference was observed between the autistic 5-year-olds and the autistic 3-year-olds in either of the three language conditions (see the right panel of Fig. 5).

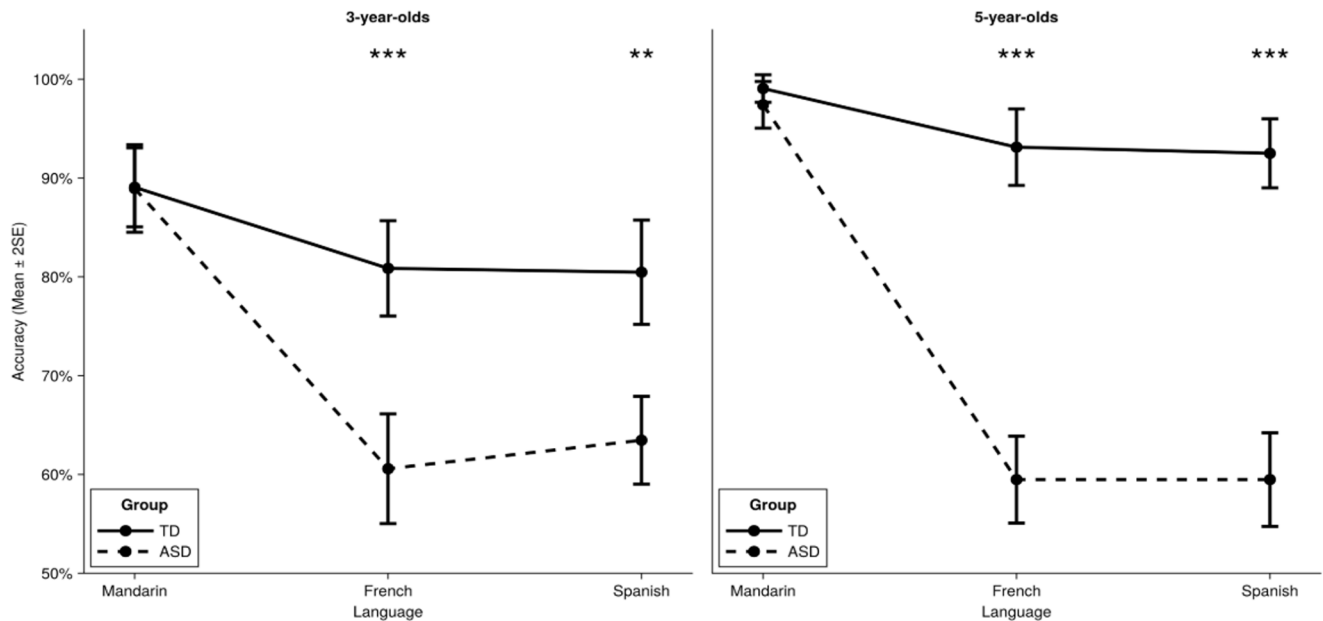


Fig. 4 The accuracy differences between the autistic and TD children in each age group (3- and 5-year-olds) under each language condition (Mandarin, French and Spanish). $**p < .01$, $***p < .001$, Bonferroni corrected

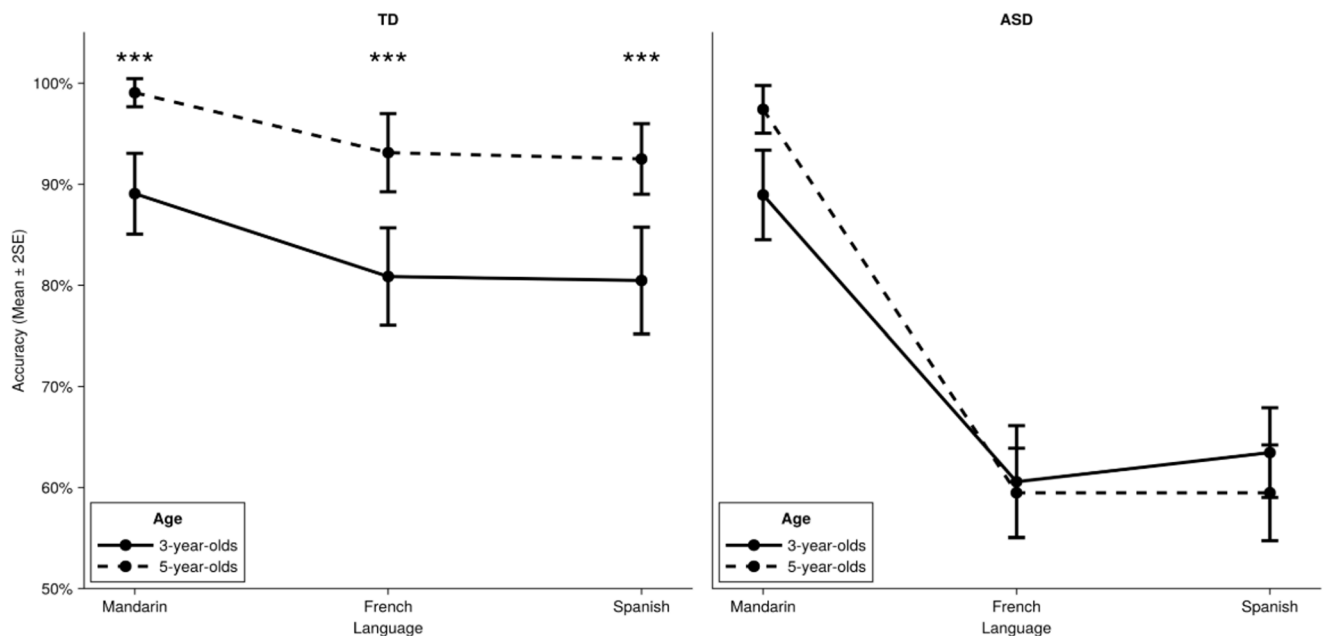


Fig. 5 The accuracy differences between the two age groups (3- and 5-year-olds) in both the autistic group (right panel) and the TD group (left panel) under each language condition (Mandarin, French and Spanish). $***p < .001$, Bonferroni corrected

Third, the comparisons between different language conditions in each group were presented in Fig. 6. As shown in the figure, no significant differences were observed in the three language conditions for TD-3-year-olds. But a significant difference was observed for the TD 5-year-olds. They performed significantly better in the Mandarin condition than in the two foreign language conditions, but no significant difference was observed between the two foreign

language conditions. The autistic 5-year-olds and the autistic 3-year-olds exhibited similar patterns. Their accuracy in the Mandarin condition was significantly higher than that in the French and Spanish conditions, and no difference was observed between the two foreign language conditions.

In addition, to better understand the difference between the autistic children and their verbal-IQ-matched TD peers, we compared the response patterns between the autistic

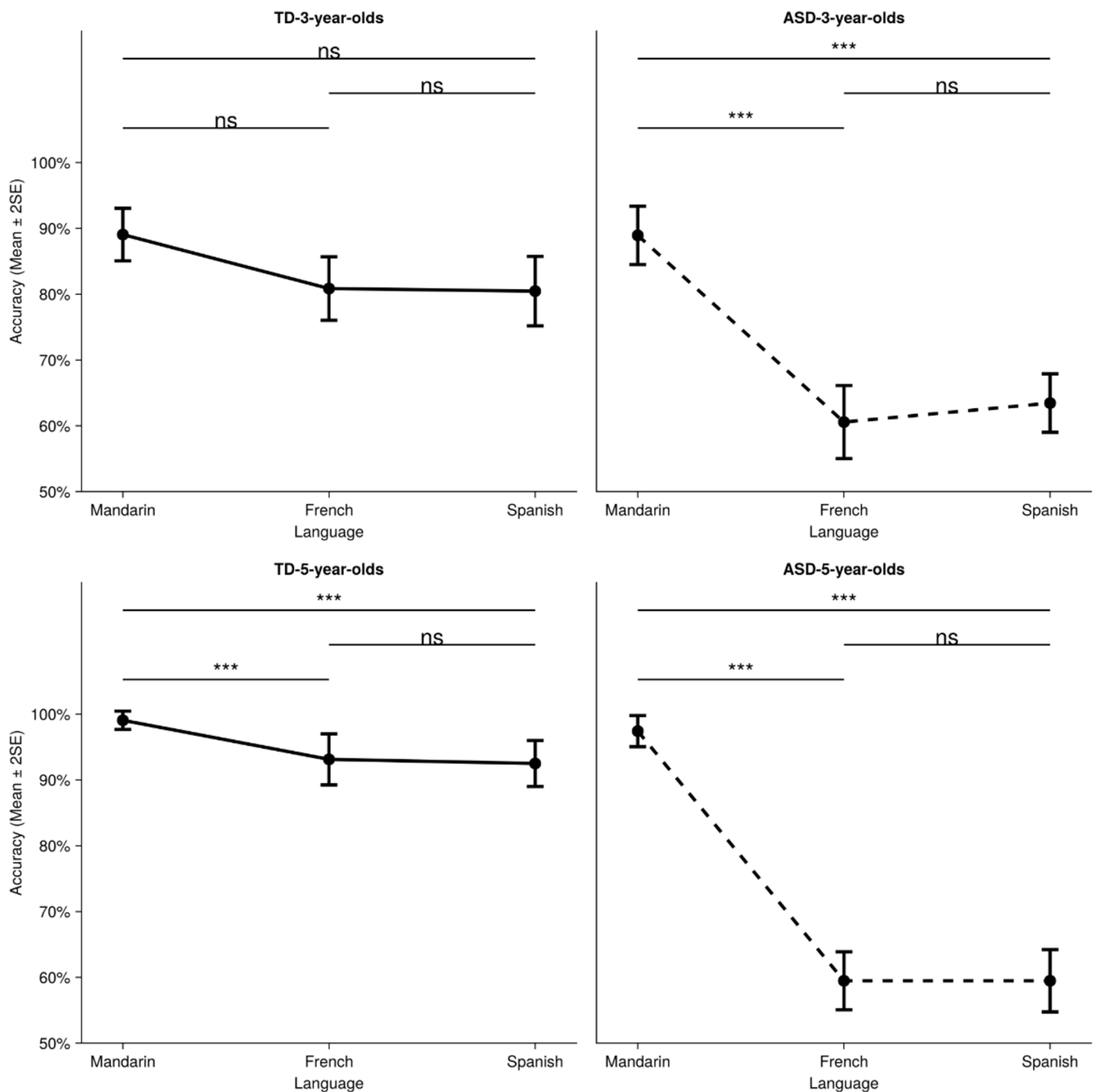


Fig. 6 The accuracy differences between the three languages (Mandarin Chinese, French, and Spanish) in each group. *** $p < .001$, Bonferroni corrected

5-year-olds and the TD 3-year-olds. As discussed in the *Participants* section, these two groups were matched on verbal IQs. The maximal model was $accuracy \sim 1 + IQ\text{-Matched-Group} * Language + (1 + Language | Subject)$ in *Julia*. The simplest best-fitting model was $accuracy \sim 1 + IQ\text{-Matched-Group} * Language + (1 | Subject)$. The results of the simplest best-fitting model are summarized in Table 5. The bootstrapping results of the observed effects were provided in Fig. 7. As indicated in the figure, all the significant

effects observed in the original model fitting were robust in our simulated samples. The results of the post-hoc analysis were given in Fig. 8. The results show that the autistic 5-year-olds provided significantly more accurate responses than IQ-matched TD 3-year-olds when the test language was Mandarin, but their accuracy was significantly lower than TD 3-year-olds when the test language was French or Spanish.

Table 5 Statistical results of the simplest best-fitting model for the two IQ-matched groups

Fixed effect	β	SE	z	Pr(> z)
(Intercept)	3.69	0.43	8.66	<1e-17***
IQMatched: TD 3-year-olds	-1.54	0.47	-3.24	0.0012**
Language: French	-3.29	0.44	-7.42	<1e-12***
Language: Spanish	-3.29	0.44	-7.42	<1e-12***
IQMatched: TD 3-year-olds & Language: French	2.62	0.51	5.11	<1e-06**
IQMatched: TD 3-year-olds & Language: Spanish	2.60	0.51	5.06	<1e-16***

** $p < .01$, *** $p < .001$

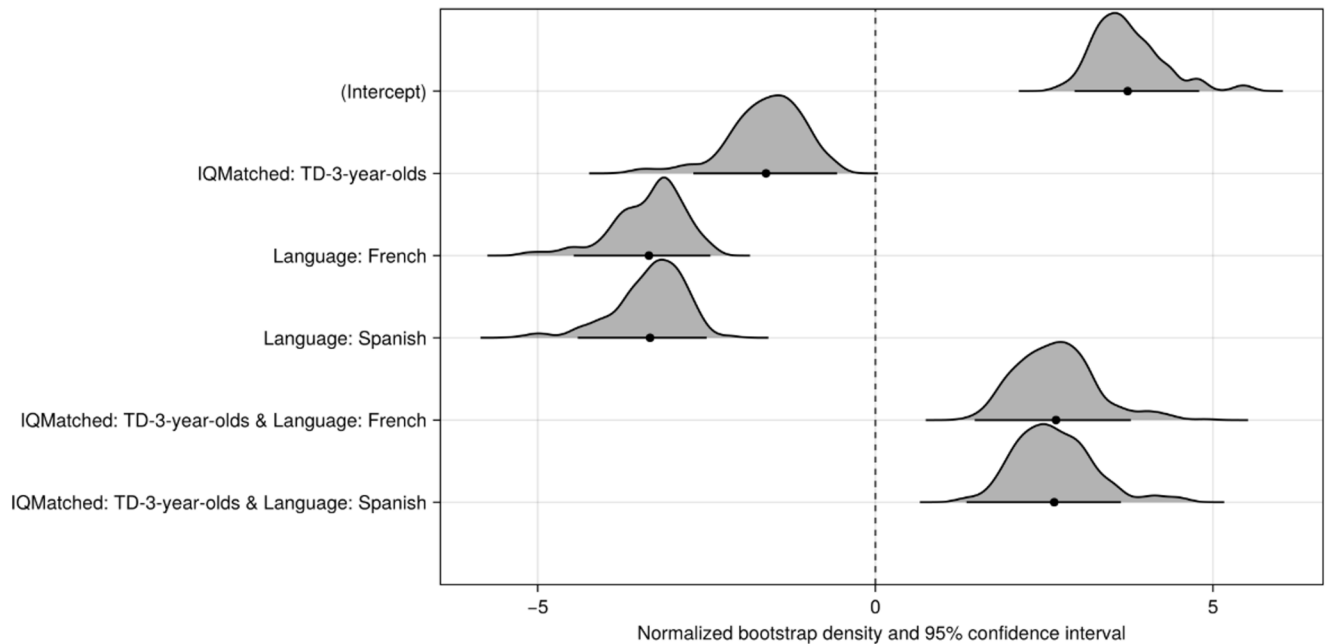


Fig. 7 The ridge plot for the bootstrap samples of the fixed effects for the simplest best-fitting model to the IQ-matched data. Each row in the figure corresponds to the results of one specific fixed effect, consisting of the original estimated coefficient (the dot), the 95% confidence

interval (the line), and the density plot of the simulated values. The density plot and/or the confidence interval that do not contain the zero value suggests that the original estimate of the fixed effect was significantly robust

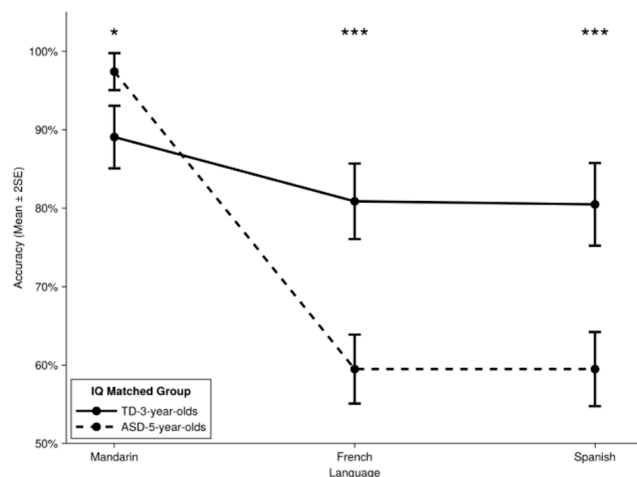


Fig. 8 The accuracy differences between two IQ-matched groups (TD-3-year-olds vs. ASD-5-year-olds) under each test language. * $p < .05$, *** $p < .001$, Bonferroni corrected

To further examine whether the children in each condition demonstrated reliable ability in processing emotional prosody, we conducted additional analyses comparing the accuracy of each participant group in each language condition against chance level (50%). Specifically, for each group and each language condition, we fitted a binomial mixed-effects model “accuracy ~ 1 + (1 | Subject)”, and the results are summarized in Table 6. In the table, if the fitted coefficient in the “Intercept” was significantly larger than zero, this means that the accuracy of this group in this condition was significantly above chance. The results showed that the TD 3- and 5-year-olds performed significantly above chance in all three language conditions (all $ps < 0.01$). Crucially, autistic children in both age groups also showed above-chance performance not only in their native language but also in the two foreign language conditions (all $ps < 0.01$).

Table 6 Accuracy relative to chance level (50%) by participant group and language condition

Language	Group	Name	Coef.	Std. Error	z	Pr(> z)
Mandarin	TD 3-year-olds	(Intercept)	2.18	0.21	10.19	2.26311e-24***
	TD 5-year-olds	(Intercept)	8.87	2.90	3.05	0.0023**
	Autistic 3-year-olds	(Intercept)	2.14	0.23	9.26	2.11015e-20***
	Autistic 5-year-olds	(Intercept)	6.25	1.14	5.47	4.55391e-8***
French	TD 3-year-olds	(Intercept)	1.44	0.16	9.07	1.19272e-19***
	TD 5-year-olds	(Intercept)	6.55	1.07	6.13	8.86236e-10***
	Autistic 3-year-olds	(Intercept)	0.43	0.14	3.03	0.0025**
	Autistic 5-year-olds	(Intercept)	0.38	0.13	2.87	0.0041**
Spanish	TD 3-year-olds	(Intercept)	1.49	0.17	8.64	5.74789e-18***
	TD 5-year-olds	(Intercept)	3.75	0.39	9.68	3.85019e-22***
	Autistic 3-year-olds	(Intercept)	0.55	0.14	3.83	0.0001***
	Autistic 5-year-olds	(Intercept)	0.38	0.13	2.87	0.0041**

* $p < .05$, ** $p < .01$, *** $p < .001$

Discussion

Previous studies found that TD children at the age of 3 start to reliably understand emotional prosody both in their native language and in foreign languages (e.g., Baltaxe, 1991; Ma et al., 2022; Stifter & Fox, 1986), suggesting that young TD children have already established abstract mapping relations between prosody and emotion that can be applied across languages. However, it remains unclear whether autistic children demonstrate a similar pattern. Given that autistic children often experience difficulties in generalizing learned knowledge across contexts, it is highly likely that their understanding of emotional prosody is more restricted to their native language. To address this question, the present study examined Mandarin-speaking autistic and TD children's understanding of two basic emotional prosody, happy and sad, across different languages, including Mandarin to which the autistic children had sufficient exposure, and French and Spanish to which they had no exposure. The major findings were as follows.

First, the autistic 3-year-olds and 5-year-olds exhibited similar response patterns to their age-matched TD peers when responding to emotional prosody in their native language, Mandarin (see Fig. 4). But their accuracy was significantly lower than their age-matched TD peers when responding to emotional prosody in foreign languages (French and Spanish). Importantly, however, autistic children's performance in all language conditions, including the two foreign languages, was significantly above chance. Second, the TD children exhibited a clear developmental pattern, i.e., the ability to respond to emotional prosody in both native and foreign languages improved significantly from 3 years of age to 5 years of age. However, no such development was observed from the autistic 3-year-olds to the autistic 5-year-olds (see Fig. 5). Further within-group analysis showed that both the autistic children and the TD 5-year-olds performed better in the native language condition than

in the two foreign language conditions (Fig. 6). In addition, the autistic 5-year-olds showed significantly better performance than the younger TD children (TD 3-year-olds) that were matched on verbal IQ when responding to emotional prosody in their native language but showed significantly lower accuracy when the emotional prosody was presented in foreign languages (see Fig. 8). Together, these findings suggest that autistic children have the basic ability to process emotional prosody across different languages, and they show a clear native language advantage. However, their performance in unfamiliar languages is less robust and shows limited developmental change compared to that of TD children.

We now discuss the implications of the findings for understanding the nature of autistic children's processing of emotional prosody. The finding that the autistic children responded to emotional prosody in their native language equally well as age-matched TD children suggests that emotional prosody processing is not an inherent difficulty in autism.² But at the same time, the autistic children responded to emotional prosody in foreign languages significantly less accurately than both the age-matched and verbal-IQ-matched TD children, although their performance remained above chance. Note that both the autistic and TD children had no exposure to French or Spanish, yet the TD children were able to correctly choose the facial expressions that matched the corresponding emotional prosody, whereas the autistic children were comparatively less able to do so.

These patterns observed in autistic children invite us to think about the nature of their mapping between prosody and emotion across languages. Prior research on TD children has suggested that this mapping might be cross-linguistic,

² As pointed out by an anonymous reviewer, this interpretation should be made with caution, as the present task involved highly exaggerated emotional contrasts, and thus it remains unclear whether a similar pattern would be observed for more subtle or complex prosody-emotion mappings.

i.e., children acquire the abstract knowledge of emotional prosody relatively early in development and apply this knowledge when decoding emotional prosody in both familiar and unfamiliar languages, without relying on transfer from the native language *per se* (Ma et al., 2022; Nagels et al., 2020). For example, Nagels et al. (2020) reported no significant differences between English- and Dutch-speaking 4- to 6-year-old TD children in their understanding of emotional prosody expressed in Dutch. In contrast, an alternative view holds that children might establish the abstract mapping between prosody and emotion early based on their native language, and this abstract knowledge of emotional prosody enables them to generalize inferences to unfamiliar languages. This is supported by previous evidence that 8- to 10-year-old English-speaking children showed higher accuracy for emotional prosody understanding in their native language than in unfamiliar foreign languages (Chronaki et al., 2018). According to this view, extensive exposure to emotional prosody in one's native language is important for children to establish the abstract mapping and then generalize it to other languages.

Autistic children's response patterns appear to be more consistent with the second account. Specifically, they performed comparably to their age-matched TD peers when responding to emotional prosody in their native language, but their accuracy was significantly lower than that of both age-matched and verbal IQ-matched TD peers when responding to emotional prosody in foreign languages. This effect of language familiarity suggests that autistic children are actually able to establish the abstract mapping relation between prosody and emotion based on the language being spoken in their daily life, and they can use this knowledge to decode the emotional states from prosodic cues embedded in the utterances that they might not have encountered before in their native language. However, unlike TD children, this mapping appears to be less applicable across different languages. Autistic children may need to rely on the mapping relations established in their native language and extend them to unfamiliar linguistic contexts. But they often face challenges in generalization, and thus the efficiency of this generalization process might be limited. This speculation is consistent with broader evidence indicating difficulties with generalization in autistic children (Church et al., 2015; de Marchena et al., 2015; Gastgeb & Strauss, 2012; Gunning et al., 2019; Naigles et al., 2013; Rincover & Koegel, 1975). Compared to TD children, autistic children

rely more heavily on sufficient language exposure to a specific language to understand the emotional prosody in that language.³

Alternatively, this effect of language familiarity might also relate to autistic children's responses to novel stimuli, i.e., sentences spoken in the languages they were not familiar with. On the one hand, it is possible that they were not interested in unfamiliar linguistic stimuli that do not have meaning to them, and thus they were not willing to attend to the spoken sentences in the two unfamiliar languages, French and Spanish. To explore this possibility, 20 participants from the autistic 5-year-olds and 20 from the TD 5-year-olds who had participated in the study were invited to take part in a post-study survey. In the survey, they were asked which sound they were more interested in by choosing between a Mandarin spoken sentence and a spoken sentence in a foreign language. The spoken sentences were selected from the test sentences used in the study. Each participant heard four Mandarin sentences, two French sentences and two Spanish sentences. Each Mandarin sentence was paired with either a French sentence or a Spanish sentence. The TD 5-year-olds' mean proportion of choosing the Mandarin sentences as more interesting was 55.33% ($SD=10.13\%$), and the autistic 5-year-olds' mean proportion of choosing the Mandarin sentences as more interesting was 53.67% ($SD=12.08\%$). No significant difference was observed between the two groups ($p > .05$). In addition, for both groups the mean proportion of selecting the Mandarin sentences as more interesting was not significantly different from chance (both $ps > 0.05$). The findings of the post-study survey suggest that the effect of language familiarity in the autistic children cannot be simply attributed to their unwillingness to attend to unfamiliar languages.

However, we also wish to note that even if autistic children were equally interested in and willing to attend to both familiar and unfamiliar languages, their processing of novel auditory stimuli may still differ from that of familiar input, as suggested by previous studies⁴ (e.g., Maes et al., 2011; Sokhadze et al., 2009, 2017). For example, Sokhadze et al. (2009) examined autistic children's and adults' responses to novel stimuli by using an oddball paradigm and evaluating their attention-related ERPs. The results revealed that the autistic group showed slower response times and prolonged latencies of attention-related ERP components than the TD controls. The findings of these prior studies raise the possibility that the autistic children in the current study were

³ As noted by an anonymous reviewer, language exposure is vital for all children's prosody-emotion mapping, not just for autistic children. What we wish to highlight here is the role that exposure to the language being spoken plays in autistic children's understanding of emotional prosody.

⁴ We thank an anonymous reviewer for pointing out this possibility.

slower than their TD peers in orienting their attention to and processing the novel auditory stimuli presented in foreign languages, and thus they were less efficient in decoding the embedded emotional information. This explains why autistic children performed above chance when responding to emotional prosody in foreign languages, yet their accuracy rates were significantly lower than their TD peers. It might be possible that their performance in the two foreign language conditions would be improved if they were given more time and explicit instructions to process the novel stimuli.

Interestingly, the native language advantage observed in the autistic children was also found in the TD 5-year-olds, but not in the TD 3-year-olds. It is important to note, however, that the nature of this advantage might be different between the two groups. For the TD 5-year-olds, the accuracy rates were above 92% across languages, and their performance in their native language, Mandarin, approached ceiling. By contrast, although the accuracy rates for Mandarin were also high in the autistic groups (89.37% for the 3-year-olds and 97.28% for the 5-year-olds), their performance in the two foreign language conditions was significantly lower, which was just above chance level. The comparison between these two groups suggests that TD children might be able to map prosody onto emotion across languages relatively early in development and that with increasing exposure to their native language, a language-specific advantage becomes more apparent. In autistic children, however, the mapping between prosody and emotion might be more strongly tied to the language to which they are exposed, and they have some challenges when applying this knowledge to unfamiliar languages, which leads to their native language advantage.

In addition, it is also worth noting that the TD children exhibited a developmental pattern in the ability to respond to emotional prosody in both native and foreign languages; however, no such development was observed in the autistic children, indicating that there was no development of this ability from 3 to 5 years of age in autistic children. But the possibility remains that development of this ability may proceed beyond 5 years of age and might occur at a rather late stage. Further research is necessary to explore whether there is indeed a late development of this ability. Longitudinal studies would be particularly ideal to observe the developmental changes in autistic children.

Finally, the differences between autistic and TD children discussed above led us to further think about whether cross-linguistic emotional prosody processing relies on the same underlying mechanisms in the two groups. One possibility is that autistic children differ from their TD peers in how the mapping relation between prosody and emotion is established and applied. TD children may acquire general knowledge about this mapping early in development that can be

applied to both familiar and unfamiliar languages without requiring the transfer from native to foreign languages. In contrast, autistic children might rely more heavily on sufficient exposure to their native language in order to establish the mapping relation and then extend this knowledge to unfamiliar languages, which is constrained by their ability to make generalizations across contexts. An alternative possibility is that the underlying mechanisms of the two groups are quite similar. Like TD children, autistic children also acquire abstract knowledge about mapping relations early on. However, they need to use it with their familiar language, as they often face challenges in processing unfamiliar stimuli. Both interpretations are consistent with the response patterns of the autistic children in the current study and previous literature about autistic children's neurocognitive profiles. In fact, we believe that understanding whether the underlying mechanisms of autistic children are identical to those of TD children may be less informative than figuring out under which conditions autistic children show strengths and relative weakness in the processing of emotional prosody. From a neurodiversity-informed perspective, clarifying these questions has more practical value for tailoring communication strategies and supports for autistic children.

Several directions for future research follow from the present findings. First, the current study relied on an explicit, offline pointing task to assess autistic children's understanding of emotional prosody. Prior research has suggested that implicit understanding of vocal and facial emotion might precede explicit behavioural responses (Berman et al., 2016; San Juan et al., 2017), which raises the possibility that autistic children may show more robust implicit sensitivity to emotional prosody than is captured by an offline pointing task. Future studies using online methods such as eye tracking could therefore provide a more fine-grained characterization of emotional prosody processing in autistic individuals. Second, the present study only focused on two basic emotions, happiness and sadness, which were presented with relatively clear and exaggerated prosodic contrasts. Thus, it remains unclear whether autistic children would show similar performance in decoding more subtle or perceptually/conceptually more complex emotional prosodies. A related issue is that both the audio and visual stimuli used in the present study were prepared by neurotypical individuals, and thus children were required to decode neurotypical emotional prosody and map it onto neurotypical facial expressions. Previous research has suggested that autistic adults differ from their TD peers in both the production and perception of emotional prosody and that they are more accurate in identifying autistic than neurotypical emotional prosody (Hubbard et al., 2017). Further studies are needed to examine how autistic children map a broader

range of autistic emotional prosody to autistic emotional faces in both native and foreign languages, which would provide a fairer and more comprehensive understanding of emotional prosody processing in autism. Finally, the experimental stimuli in the present study consisted of isolated emotional prosodic expressions paired with line-drawn schematic emotional faces. Future studies could examine emotional prosody processing in more naturalistic settings, such as conversational or interactive tasks, and incorporate more naturalistic facial stimuli. Importantly, such studies should include autistic communicative contexts, rather than focusing exclusively on neurotypical contexts.

The present findings also have potential implications for developing supportive strategies targeting emotional communication in autistic children. Importantly, autistic children demonstrated above-chance performance across all language conditions and showed particularly strong sensitivity to emotional prosody in their native language, indicating preserved foundational ability to process emotional prosody. Within a neurodiversity framework, these strengths may be productively leveraged to support further development, rather than framing emotional prosody understanding as a domain of deficit. One promising direction is to focus on supporting generalization by building on children's robust prosody-emotion mappings in their native language. For example, activities could incorporate explicit cross-linguistic comparisons, in which emotional prosody in the child's native language is presented alongside corresponding prosodic expressions in an unfamiliar language, accompanied by congruent emotional facial cues. Such comparisons may help highlight shared prosodic features of emotions across languages (e.g., higher pitch and faster tempo for happiness), thereby fostering more abstract representations of emotional prosody.

Overall, our findings indicate that autistic children are able to decode emotional prosody not only in their native language but also in foreign languages, although their ability to process emotional prosody in foreign languages is less developed than their TD peers. This pattern might point to fundamentally different mechanisms between autistic and TD children in how prosody-emotion mappings are acquired and applied. Alternatively, it is also possible that both groups share the same underlying mechanisms, but some autism-related neurocognitive characteristics constrain autistic children's efficiency of processing emotional prosody in unfamiliar languages. Within the autistic group, we also observe a native language advantage, and this advantage can be leveraged to provide supports targeting emotional communication in autistic children. Taken together, the present study provides a cross-linguistic perspective on autistic children's processing of emotional prosody and sheds light on our understanding of their neurocognitive profiles.

Author Contributions This study was conceptualised by P.Z., W.M. and X.Z., with contributions from L.Z. and J.S. Test stimuli were constructed by P.Z. and W.M. Data were collected and analysed by P.Z. and L.Z. Manuscript was drafted and revised by P.Z. and was approved by all authors before submission. P.Z., W.M. and X.Z. contributed equally to this study.

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Data Availability The data that support the findings of this study, including the test stimuli, the raw data, the Julia code, and the modal fitting outputs, are available in Open Science Framework via the link: https://osf.io/9u4yp/?view_only=f53ccf81f9b8400c897df740b65a1398.

Declarations

Conflict of interest The authors declare no conflicts of interest.

Ethical Approval The study was approved by the Ethics Committee of the School of International Studies, Zhejiang University, SIS2023-04. Written informed consent has been obtained from each participant's parent.

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